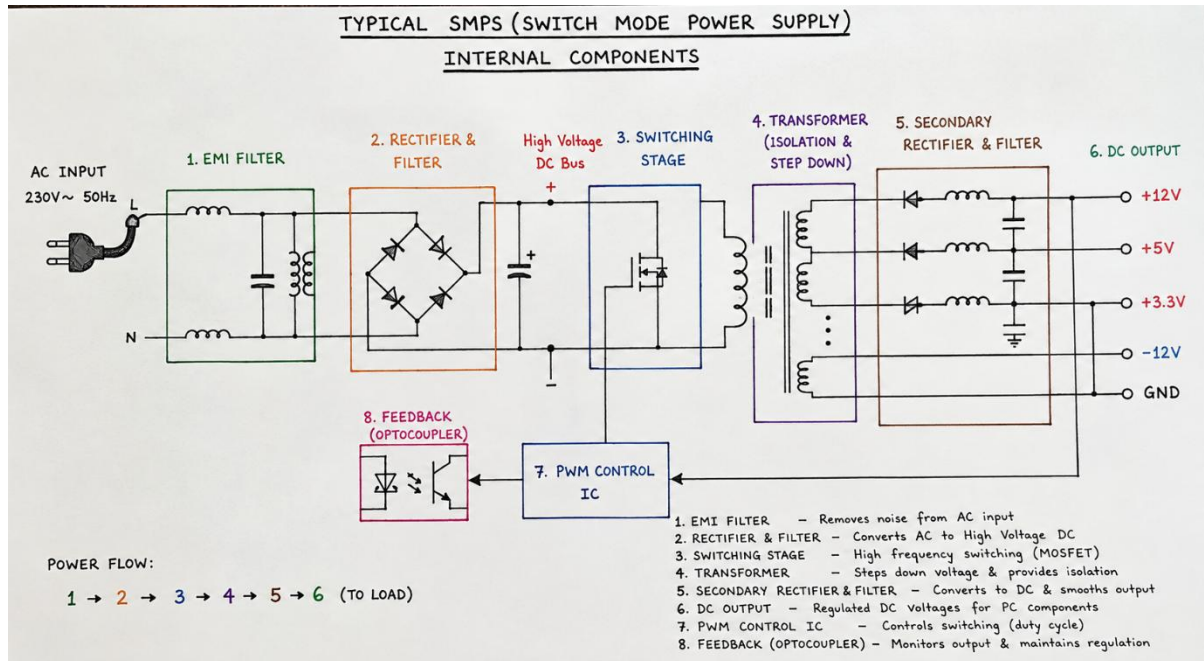


1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.



Device	Type	Why it needs this power supply
Desktop Computer	SMPS	Converts AC power into the correct DC voltages needed by computer components.
LED TV	SMPS	Provides stable and efficient power to the TV's electronic circuits.
Wi-Fi Router	SMPS	Converts household AC power into low-voltage DC power for the router.
Desktop Computer (with UPS)	UPS	Keeps the computer running for a short time during a power cut and prevents data loss.
CCTV system	UPS	Provides backup power so cameras and recording continue working during power failures.

2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections. You can hand-draw and upload a photo, or use any free diagram tool like draw.



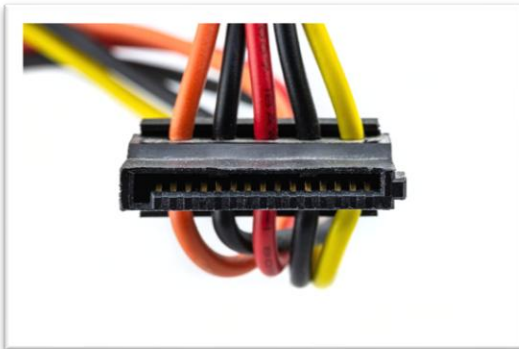
3.Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.



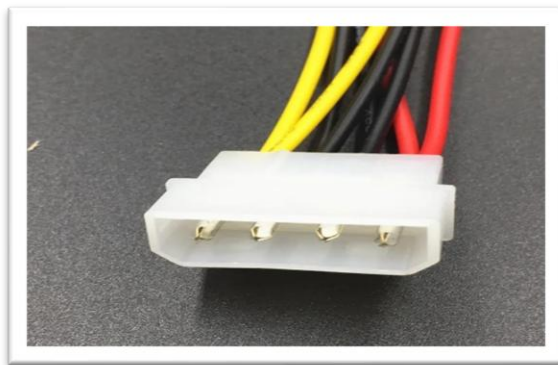
1) ATX 24 pin motherboard connectors:



2) SATA power connectors:



3) Molex connectors:



4) PCIe 6/8-pin connector.



Power Connectors	Device It Powers	Safety Tips
ATX 24 pins	Powers the motherboard	Turn off and unplug the PC before connecting or removing it
SATA Power	Powers HDDs, SSDs, and SATA drives.	Do not force the connector; align it correctly before plugging in.
Molex	Powers older HDDs, fans, and some accessories.	Hold the connectors firmly and pull it straight out without pulling the wires
PCIe 6/8 pin	Provide extra power to a graphics card (GPU)	Make sure the correct PCIe cable is connected before turning on the PC

4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.



1) Install the Power Supply:

Keep the power switch on the back of the power supply set to OFF (0).

Place the power supply box into the bottom of your PC case with its fan facing down toward the floor vents.

2) Power the Motherboard:

Cable: Thick 24-Pin Cable (the widest plug coming from the power supply).

Find the large socket on the right edge of the motherboard. Push the wide 24-pin plug into it until the plastic clip clicks locked.

3) Power the Processor (CPU):

Cable: 8-Pin CPU Cable (often splits into two 4-pin halves, usually labeled CPU).

Find the socket at the top-left corner of the motherboard. Plug in the 8-pin CPU cable until it clicks into place.

4) Power the Graphics Card:

Cable: 6+2 Pin PCIe Cable (usually labeled PCIe or GPU).

Plug the PCIe cable into the power ports on the side of your graphics card. Make sure it is pushed all the way in so there are no gaps.

5) Power Hard Drives & SSDs:

Cable: Flat, L-Shaped SATA Power Cable.

Plug the flat SATA cable into the back of any standard 2.5-inch SSD or 3.5

